

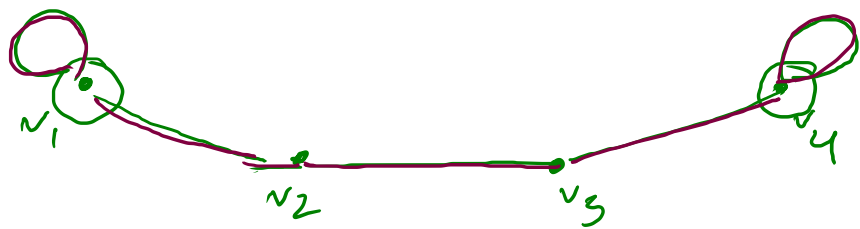
10.5 Euler + Hamiltonian paths.

① Theorem 1: A connected multigraph has an Euler path not an Euler circuit iff there is exactly 2 vertices of odd degree.

② Theorem 2: A connected multigraph with at least 2 vertices has an Euler circuit iff each vertex has even degree.

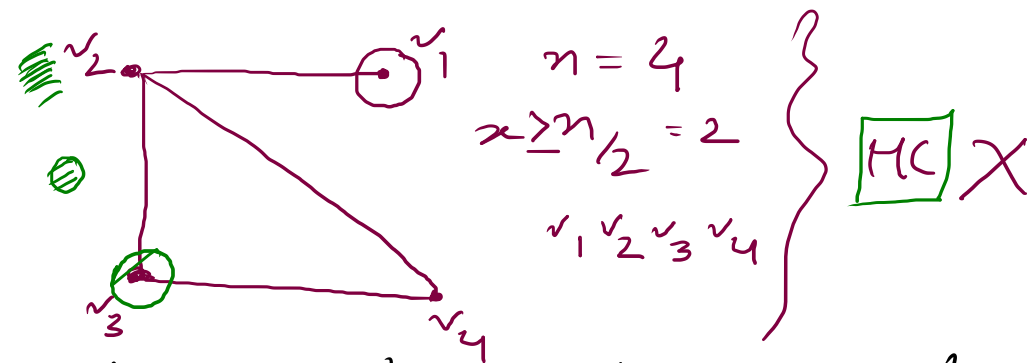
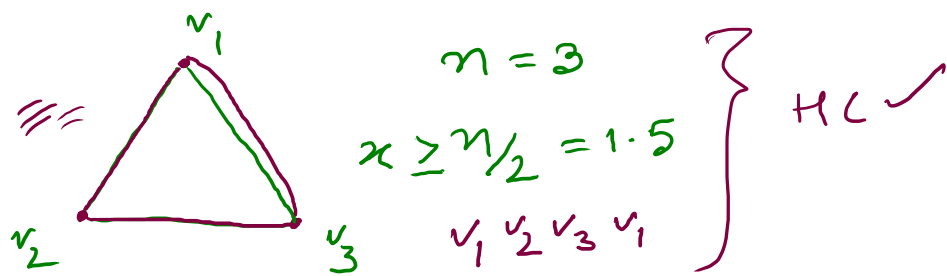
$$d_{v_1} = d_{v_2} = 2 \rightarrow \text{even.}$$

E. circuit $\rightarrow v_1 v_2 v_1 \rightarrow \text{Th}_2$.



$$\left. \begin{array}{l} d_{v_1} = d_{v_4} = 3 \rightarrow \text{odd.} \\ v_1 v_2 v_3 v_4 v_4 \rightarrow \text{EP.} \end{array} \right\} \rightarrow \text{Th}_1$$

Dirac's theorem: If G is a simple graph with n vertices and $n \geq 3$ such that degree of each vertex is at least $n/2$ then G will have HC.



Ore's theorem: If G is a simple graph, with n vertices and $n \geq 3$ such that for every pair of non-adjacent vertices $(u, v) \rightarrow \deg(u) + \deg(v) \geq n$, then G has a HC.

